

Kubernetes Lab Questions

These exercises provide practical experience with key Kubernetes concepts and operations.

1. Installing Minikube on Ubuntu VM

- Install Minikube on an Ubuntu virtual machine, verify the Minikube version, and access the Kubernetes dashboard.

2. Namespace Management

- Using appropriate commands, create a demo-namespace, list all namespaces, and delete the demo-namespace.
- Create a namespace using a namespace.yaml configuration file.

3. Practicing Basic Kubernetes Commands

- Create nginx-deployment.yaml and nginx-service.yaml files.
- Apply the configurations.
- List all pods, services, and deployments.
- Describe all pods, services, and deployments.

4. Persistent Storage with PV and PVC

- Write YAML files to define PersistentVolume (PV) and PersistentVolumeClaim (PVC).
- Deploy a pod using PVC and verify persistent storage.
- Validate data persistence after deleting Kubernetes resources.

5. Pod-to-Pod Communication

- Create and apply backend-pod.yaml and frontend-pod.yaml.
- Create and apply backend-service.yaml.
- Verify frontend logs and access the application inside the pod using port-forwarding.
- Access the backend using the command: `curl http://localhost:8080`.

6. Deploying ClusterIP Service

- Write necessary nginx-deployment and nginx-clusterip-service YAML files.
- Apply the configurations and verify the ClusterIP service using a pod with curl.

7. Deploying NodePort Service

- Write an nginx-nodeport-service YAML file.
- Apply the configuration and access the Nginx server using the allocated static port number.

8. Implementing Ingress Controller

- Write an nginx-ingress YAML file.
- Apply the configuration and access the server using the nginx.local URL.

9. ConfigMaps in Kubernetes

- Write a configmap.yaml file and assign key-value pairs.
- Access the ConfigMap using a pod-with-configmap.yaml file.
- Apply the configurations and print the key value inside the pod.

10. Managing Secrets in Kubernetes

- Write a secret.yaml file to store base64 encoded values for a username and password.
- Access the secret variables using a pod-with-secret.yaml file.
- Apply the configurations and print the username and password.

11. Demonstrating Live Updates

- Create a ConfigMap, deployment, and service for a Nginx server.
- Access the application via minikubeIP:port.
- Modify the ConfigMap, apply the changes, refresh the browser, and confirm that updates are reflected without restarting the service.

12. Resource Quotas in Kubernetes

- Write a resource quota configuration file for the dev-environment namespace and verify the applied resource limits.

13. Defining Resource Limits

- Write a resource limits configuration file for a pod.
- Apply the configuration and verify the limits.

14. Horizontal Pod Autoscaler (HPA) Demonstration

- Write nginx-deployment.yaml, nginx-service.yaml, and nginx-hpa.yaml files.
- Apply the configurations.
- Write a system load testing script and execute it.
- Verify the Horizontal Pod Autoscaler's scaling behavior under load.